

CALCULUS C: REVIEW OF THE MIDTERM

WEI WANG

1. CALCULATING THE LIMIT

1.1. Sandwich theorem. Recall that if $a_n \leq b_n \leq c_n$ and $a_n, c_n \rightarrow x$, then $b_n \rightarrow x$. For limit of functions, if $f(x) \leq g(x) \leq h(x)$ and

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} h(x) = A,$$

then

$$\lim_{x \rightarrow x_0} g(x) = A$$

Example 1.1. Assume that $0 < a_1 \leq a_2 \leq \dots \leq a_k$. Calculate

$$\lim_{n \rightarrow +\infty} (a_1^n + \dots + a_k^n)^{\frac{1}{n}}.$$

Proof. We see that

$$a_k \leq (a_1^n + \dots + a_k^n)^{\frac{1}{n}} \leq k^{\frac{1}{n}} a_k.$$

Then $\lim_{n \rightarrow +\infty} (a_1^n + \dots + a_k^n)^{\frac{1}{n}} = a_k$. □

Exercise 1.2. Assume that $0 < a_1 \leq a_2 \leq \dots \leq a_k$. Calculate

$$\lim_{n \rightarrow +\infty} (a_1^{\frac{1}{n}} + \dots + a_k^{\frac{1}{n}})^n.$$

Exercise 1.3. Assume that $a_k \rightarrow a > 0$. Calculate

$$\lim_{n \rightarrow +\infty} (a_1^n + \dots + a_n^n)^{\frac{1}{n}}.$$

1.2. Cauchy's proposition. It is about the limit of average for a sequence having a limit.

Proposition 1.4. If $\lim_{n \rightarrow +\infty} x_n = x$, then

$$\lim_{n \rightarrow +\infty} \frac{\sum_{i=1}^n x_i}{n} = x. \tag{1.1}$$

Proof. For any $\varepsilon > 0$, we can choose $N \in \mathbb{Z}_+$ such that $|x_n - x| < \varepsilon$ for any $n > N$. This implies that if $n > N$, there holds

$$\left| \frac{\sum_{i=1}^n x_i}{n} - x \right| \leq \frac{\sum_{i=1}^N |x_i|}{n} + \frac{n - N}{n} \varepsilon.$$

Choosing larger N without changing the notation, we can assume that if $n > N$, then

$$\left| \frac{\sum_{i=1}^n x_i}{n} - x \right| < 2\varepsilon.$$

As a result, we have (1.1). □

1.3. Comparing the order. Let $a > 0$ and $b > 1$, we have

$$\ln n \ll n^a \ll b^n \ll n! \ll n^n, \quad (1.2)$$

where we call $f(n) \ll g(n)$ if

$$\lim_{n \rightarrow +\infty} \frac{f(n)}{g(n)} = 0.$$

Now, let us show (1.2). We note that

$$0 < \frac{n!}{n^n} \leq \frac{1}{n},$$

which implies that $n! \ll n^n$. Let $x_n = b^n/(n!)$. We have

$$\frac{b_{n+1}}{b_n} = \frac{b}{n+1}.$$

As a result, $b^n \ll n!$ follows from the following lemma, whose proof is left for the reader.

Lemma 1.5. *Let $\{x_n\}$ be a sequence, if*

$$\lim_{n \rightarrow +\infty} \left| \frac{x_{n+1}}{x_n} \right| < 1,$$

then $\lim_{n \rightarrow +\infty} x_n = 0$.

Denote $y_n = \frac{n^a}{b^n}$. By using this lemma, we obtain that $n^a \ll b^n$. Finally, we turn to the proof of $\ln n \ll n^a$. Noting that

$$\frac{\ln n}{n^a} = \frac{1}{a} \cdot \frac{\ln n^a}{n^a},$$

we only show that $\frac{\ln n}{n} \rightarrow 0^+$. It is a direct consequence of the following exercise.

Exercise 1.6. $\lim_{n \rightarrow +\infty} n^{\frac{1}{n}} = 1$.

Exercise 1.7 (2024 Fall Mid). Calculate

$$\lim_{n \rightarrow +\infty} \sqrt[n]{100 + \frac{1}{n}}.$$

Exercise 1.8 (2023 Fall Mid). Calculate

$$\lim_{n \rightarrow +\infty} \sqrt[n]{\ln n}.$$

Exercise 1.9 (2023 Fall Mid). Calculate

$$\lim_{n \rightarrow +\infty} \frac{3^n}{n!}.$$

Exercise 1.10 (2022 Fall Mid). Calculate

$$\lim_{n \rightarrow +\infty} \sqrt[n]{2022 + \sin n}.$$

We also have the following equivalences:

- (1) $e^x - 1 \sim x$ ($x \rightarrow 0$);
- (2) $\sin x \sim x$ ($x \rightarrow 0$);
- (3) $\ln(1+x) \sim x$ ($x \rightarrow 0$);
- (4) $1 - \cos x \sim \frac{1}{2}x^2$ ($x \rightarrow 0$);
- (5) $(1+x)^\alpha - 1 \sim \alpha x$ ($x \rightarrow 0$);
- (6) $\arctan x \sim x$ ($x \rightarrow 0$).

Remark 1.11. The equivalences above can be only used in the calculations of limits with the fractional form.

Exercise 1.12 (2023 Fall Mid). Calculate

$$\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}.$$

Exercise 1.13 (2022 Fall Mid). Calculate

$$\lim_{x \rightarrow 1} \tan \left(\frac{\sin \pi x}{4(x-1)} \right).$$

1.4. **Limits with respect to the definition of e .** We know that

$$e = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^n. \quad (1.3)$$

By this, we can consider related limits.

Exercise 1.14 (2024 Fall Mid). Calculate

$$\lim_{x \rightarrow 0} (1 + 3x)^{\frac{1}{\sin x}}$$

Exercise 1.15 (2024 Fall Mid). Calculate

$$\lim_{x \rightarrow 1} (\sqrt{x})^{\frac{1}{\sqrt{x}-1}}.$$

Exercise 1.16 (2023 Fall Mid). Calculate

$$\lim_{x \rightarrow +\infty} \left(\frac{x^3}{(x-1)(x-2)(x-3)} \right)^x.$$

Exercise 1.17 (2022 Fall Mid). Calculate

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x^2 - 1} \right)^{x^2}.$$

Exercise 1.18 (2022 Fall Mid). Calculate

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n} - 1}{\ln \sqrt[n]{n}}.$$

1.5. L'Hôpital's rule.

Theorem 1.19 (L'Hôpital's Rule for $\frac{0}{0}$ Form). *Suppose that functions $f(x)$ and $g(x)$ satisfy the following conditions:*

- (1) $\lim_{x \rightarrow x_0} f(x) = 0$ and $\lim_{x \rightarrow x_0} g(x) = 0$;
- (2) $f(x)$ and $g(x)$ are differentiable in a deleted neighborhood of x_0 , and $g'(x) \neq 0$ in this neighborhood;
- (3) $\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = A$ (where A can be a finite number, $+\infty$ or $-\infty$).

Then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = A.$$

The rule also holds when $x \rightarrow x_0^+$, $x \rightarrow x_0^-$, $x \rightarrow +\infty$ or $x \rightarrow -\infty$.

Theorem 1.20 (L'Hôpital's Rule for $\frac{\infty}{\infty}$ Form). *Suppose that functions $f(x)$ and $g(x)$ satisfy the following conditions:*

- (1) $\lim_{x \rightarrow x_0} f(x) = \infty$ and $\lim_{x \rightarrow x_0} g(x) = \infty$;
- (2) $f(x)$ and $g(x)$ are differentiable in a deleted neighborhood of x_0 , and $g'(x) \neq 0$ in this neighborhood;
- (3) $\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = A$ (where A can be a finite number, $+\infty$ or $-\infty$).

Then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = A.$$

The rule also holds when $x \rightarrow x_0^+$, $x \rightarrow x_0^-$, $x \rightarrow +\infty$ or $x \rightarrow -\infty$.

Remark 1.21. L'Hôpital's Rule only applies to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ indeterminate forms. For other forms (e.g., $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , ∞^0), convert them to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ first.

If $\lim \frac{f'(x)}{g'(x)}$ does not exist (and is not ∞), L'Hôpital's Rule cannot be used, but the original limit may still exist.

Repeated application is allowed if the resulting limit still satisfies the conditions of the rule.

Exercise 1.22 (2022 Fall Mid). Calculate

$$\lim_{x \rightarrow 0} \frac{x - \arcsin x}{(\tan x)^2 \sin x}.$$

1.6. Taylor's expansion.

Theorem 1.23 (Taylor's expansion with Peano's remainder). Assume that f is n -differentiable at x_0 . Then

$$\begin{aligned} f(x) = & f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots \\ & + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + o((x - x_0)^n) \quad (x \rightarrow x_0) \end{aligned}$$

Theorem 1.24 (Taylor's expansion with Lagrange's remainder). Assume that f is $(n+1)$ -differentiable in $(x_0 - \delta, x_0 + \delta)$. Then for any $x \in (x_0 - \delta, x_0 + \delta)$ with $x \neq x_0$, there is ξ in the interval of x_0 and x such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + r_n(x),$$

where

$$r_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

Remark 1.25. Common Taylor expansions at $x_0 = 0$ (Maclaurin expansions) with Peano's remainder:

- (1) $e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n)$, $x \rightarrow 0$;
- (2) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + o(x^{2k+1})$, $x \rightarrow 0$;
- (3) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^k \frac{x^{2k}}{(2k)!} + o(x^{2k})$, $x \rightarrow 0$;
- (4) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$, $x \rightarrow 0$;
- (5) $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + o(x^n)$, $x \rightarrow 0$ (for any real α).

Exercise 1.26 (2024 Fall Mid). Calculate

$$\lim_{x \rightarrow 0} \frac{\cos x - \cos x^2}{x^2}.$$

2. CALCULATING THE DERIVATIVES

This section covers derivative calculation via definition, basic formulas, implicit functions, and parameterization.

2.1. Applying the definition. For $y = f(x)$,

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

Right derivative: $f'(x_0+) = \lim_{\Delta x \rightarrow 0^+} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ (left derivative similarly for $\Delta x \rightarrow 0^-$).

Differentiability: Exists iff left and right derivatives exist and are equal.

Exercise 2.1 (2024 Fall Mid). Assume that

$$f(x) = \begin{cases} \ln(1 + x^2 \sin \frac{1}{x}), & 0 < x < 1, \\ 0, & x = 0. \end{cases}$$

Consider the existences of $f(0+)$, right derivative $f'(0+)$ and $\lim_{x \rightarrow 0+0} f'(x)$.

Exercise 2.2 (2023 Fall Mid). Let $f(x)$ be differentiable at 0, and when $|x| < 1$, there is $|f(x)| \leq \ln(1 + |\arcsin x|)$. Prove that $|f'(0)| \leq 1$.

2.2. Applying basic formulae.

- **Basic formulae:** Derivatives of elementary functions (power, exponential, logarithmic, trigonometric, inverse trigonometric).
- **Rules:** Sum, product, exponent, quotient, and chain rule (for composites: $(f(g(x)))' = f'(g(x))g'(x)$).

Exercise 2.3 (2024 Fall Mid). Calculate $f'(x)$, where

$$f(x) = \arctan(1 + \sin x) + (1 + x^2)^{x^2}.$$

2.2.1. *Implicit function.* Implicit function is Defined by $F(x, y) = 0$ with y as a function of x .

Differentiation: Differentiate both sides w.r.t. x , treat y as $y(x)$, solve for y' .

Exercise 2.4 (2023 Fall Mid). Calculate the derivative of function that is determined by

$$y \arctan(x + y) + e^y \ln(1 + (x + y)^2) = 0.$$

2.3. **Parameterization.** We have the parametric equations:

$$\begin{cases} x = \varphi(t) \\ y = \psi(t). \end{cases}$$

First derivative: $\frac{dy}{dx} = \frac{\psi'(t)}{\varphi'(t)}$ ($\varphi'(t) \neq 0$).

Second derivative: $\frac{d^2y}{dx^2} = \frac{1}{\varphi'(t)} \frac{d}{dt} \left(\frac{\psi'(t)}{\varphi'(t)} \right)$.

Exercise 2.5 (2024 Fall Mid). Assume that $x = 2t - \sin t$ and $y = \cos t$. Calculate $\frac{dy}{dx}$.

Exercise 2.6 (2022 Fall Mid). Assume that $x = \varphi(t)$ and $y = \psi(t)$. Calculate $\frac{d^2y}{dx^2}$.

3. APPLICATION OF DERIVATIVES

3.1. Calculating the maximum and minimum.

- **Extreme value:** A point where the function's derivative is zero (critical point) or undefined, and the function changes from increasing to decreasing (maximum) or vice versa (minimum).
- **Global maximum/minimum:** The largest/smallest value of the function over a given domain, found by comparing values at critical points and boundary points.
- **Optimization steps:**
 - (1) Determine the objective function and constraints.
 - (2) Find critical points by solving $f'(x) = 0$ or identifying points where $f'(x)$ is undefined.
 - (3) Analyze the sign change of $f'(x)$ (first derivative test) or the sign of $f''(x)$ (second derivative test) to classify critical points.
 - (4) Evaluate the function at critical points and boundaries to find global extrema.

Exercise 3.1 (2024 Fall Mid). Consider the ellipse $L : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Calculate the maximum value of the area of triangular ABC such that $A, B, C \in L$, $AB = AC$, and BC is paralleled to the x -axis.

Exercise 3.2 (2023 Fall Mid). Find the extreme points and extreme values of the function $f(x) = x^{\frac{1}{3}}(1 - x)^{\frac{2}{3}}$.

3.2. Proving inequalities.

- **Key method using derivatives:** Construct an auxiliary function $f(x)$, then analyze its monotonicity, extreme values, or convexity to prove the inequality.
- **Common strategies:**
 - (1) **Monotonicity:** Show $f'(x) \geq 0$ (or ≤ 0) to prove $f(x) \geq f(a)$ (or $\leq f(a)$) for $x \geq a$.
 - (2) **Extreme values:** Prove the minimum (maximum) of $f(x)$ is greater (less) than or equal to 0.
 - (3) **Convexity:** Use properties of convex/concave functions (second derivative sign) to derive inequalities.

Exercise 3.3 (2024 Fall Mid). Show that

$$\left(1 + \frac{1}{x}\right)^{x+1} > e,$$

where $x > 0$.

Exercise 3.4 (2023 Fall Mid). Let $a > \ln 2 - 1$, prove that $x^2 - 2ax + 1 < e^x$ for $x > 0$.

Exercise 3.5 (2022 Fall Mid). Prove that

$$(b + a + 1) \ln(b + a + 1) - (1 + b) \ln(1 + b) > (2a + 1) \ln(2a + 1) - (1 + a) \ln(1 + a),$$

where $b > a > 0$.

4. PROPERTIES OF CONTINUOUS FUNCTIONS

4.1. Discontinuity.

Definition 4.1 (Types of Discontinuities). Let $f(x)$ be defined in a deleted neighborhood of x_0 . We classify discontinuities of $f(x)$ at x_0 as follows:

- (1) **Removable Discontinuity:** $\lim_{x \rightarrow x_0} f(x)$ exists, but either $f(x_0)$ is not defined, or $\lim_{x \rightarrow x_0} f(x) \neq f(x_0)$.
- (2) **Jump Discontinuity:** $\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x)$ both exist but are not equal.
- (3) **Infinite Discontinuity:** $\lim_{x \rightarrow x_0} f(x) = \infty$ (or $+\infty, -\infty$), or one-sided limits are infinite.
- (4) **Oscillatory Discontinuity:** $\lim_{x \rightarrow x_0} f(x)$ does not exist and is not infinite (e.g., $f(x) = \sin \frac{1}{x}$ at $x = 0$).

Removable and jump discontinuities are collectively called **first-class discontinuities**; the other two types are **second-class discontinuities**.

Exercise 4.2 (2023 Fall Mid). Let

$$f(x) = \begin{cases} x^2 \ln \left(2 + \cos \left(\frac{1}{x}\right)\right), & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Find the derivative function $f'(x)$, and ask at which points $f'(x)$ is discontinuous and what type of discontinuity they are.

4.2. Intermediate theorem.

Theorem 4.3 (Intermediate value theorem). Let $f(x)$ be continuous on the closed interval $[a, b]$, and let $f(a) \neq f(b)$. For any real number C between $f(a)$ and $f(b)$, there exists at least one point $\xi \in (a, b)$ such that $f(\xi) = C$.

Corollary 4.4 (Zero point theorem). Let $f(x)$ be continuous on $[a, b]$, and $f(a) \cdot f(b) < 0$ (i.e., $f(a)$ and $f(b)$ have opposite signs). Then there exists at least one point $\xi \in (a, b)$ such that $f(\xi) = 0$.

Exercise 4.5 (2024 Fall Mid). Show that the equation $\ln(2 + \cos x) - \frac{1}{x} = 0$ has infinite numbers of positive roots.

Exercise 4.6 (2023 Fall Mid). Prove that the equation $\left(\frac{1}{2}\right)^x + \left(\frac{3}{4}\right)^x = 1$ has only one real root.

Exercise 4.7. Show that $f(x) = \frac{\cos x}{x^2} - (\sin x)^5$ has infinite numbers of positive roots.

5. MEAN VALUE THEOREM

Theorem 5.1 (Rolle's Theorem). *Let $f(x)$ satisfy:*

- (1) *Continuous on $[a, b]$;*
- (2) *Differentiable on (a, b) ;*
- (3) $f(a) = f(b)$.

Then there exists at least one $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Theorem 5.2 (Lagrange's Mean Value Theorem). *Let $f(x)$ satisfy:*

- (1) *Continuous on $[a, b]$;*
- (2) *Differentiable on (a, b) .*

Then there exists at least one $\xi \in (a, b)$ such that

$$f(b) - f(a) = f'(\xi)(b - a).$$

The formula can also be written as $f(a + h) - f(a) = f'(a + \theta h)h$ for some $\theta \in (0, 1)$ (where $h = b - a$).

Theorem 5.3 (Cauchy's Mean Value Theorem). *Let $f(x)$ and $g(x)$ satisfy:*

- (1) *Continuous on $[a, b]$;*
- (2) *Differentiable on (a, b) ;*
- (3) $g'(x) \neq 0$ for all $x \in (a, b)$;
- (4) $g(a) \neq g(b)$.

Then there exists at least one $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

Exercise 5.4 (2024 Fall Mid). Assume that $f(x)$ is differentiable in (a, b) . If for some $x_0 \in (a, b)$, $\lim_{x \rightarrow x_0} f'(x)$ exists. Show that $\lim_{x \rightarrow x_0} f'(x) = f'(x_0)$.

Exercise 5.5 (2023 Fall Mid). Let $g(x)$ be continuous on $[a, b]$ and differentiable on (a, b) , $g(a) = 0$, and satisfy $|g'(x)| \leq \frac{1}{2(b-a)}|g(x)|, x \in (a, b)$. Prove that $g(x) = 0, x \in [a, b]$.

6. PROOF OF THE CONVERGENCE

6.1. Key Theorems for Sequence Convergence.

Theorem 6.1 (Monotone bounded theorem). *Every monotone (non-decreasing or non-increasing) and bounded sequence must converge.*

- *If $\{x_n\}$ is non-decreasing and bounded above, then $\lim_{n \rightarrow +\infty} x_n = \sup\{x_n \mid n \in \mathbb{N}\}$;*
- *If $\{x_n\}$ is non-increasing and bounded below, then $\lim_{n \rightarrow +\infty} x_n = \inf\{x_n \mid n \in \mathbb{N}\}$.*

Theorem 6.2 (Cauchy Convergence Criterion). *A sequence $\{x_n\}$ converges if and only if for any $\varepsilon > 0$, there exists a positive integer N such that for all $m, n > N$, we have $|x_m - x_n| < \varepsilon$.*

*A sequence that satisfies the above condition is called a **Cauchy sequence**.*

Remark 6.3. Common strategies for proving sequence convergence:

- (1) For recursive sequences (e.g., $x_{n+1} = f(x_n)$), use the **Monotone Bounded Theorem**: first prove monotonicity (via induction or $x_{n+1} - x_n$), then prove boundedness (via induction or inequality estimation);
- (2) For sequences where monotonicity is hard to verify, use the **Cauchy Convergence Criterion** (estimate $|x_m - x_n|$ directly, often using telescoping series or geometric series bounds);

Exercise 6.4 (2023 Fall Mid). Let $x_{n+1} = \sin x_n, n = 0, 1, 2, \dots$, where x_0 is any real number. Prove that x_n has a limit and find the limit.

Exercise 6.5 (2022 Fall Mid). Given $0 < b < a$. Let $a_0 = a, b_0 = b$. Assume that

$$a_n = \frac{a_{n-1} + b_{n-1}}{2}, \quad b_n = \sqrt{a_{n-1}b_{n-1}}, \quad n \in \mathbb{Z}_+.$$

Let $\{a_n\}$ and $\{b_n\}$ be two sequences. Show that these two sequences have the same limit.

Exercise 6.6 (2021 Fall Mid). Let $x_1 > 0$. Assume that for any $n \in \mathbb{Z}_+$,

$$x_{n+1} = \frac{2(1+x_n)}{2+x_n}.$$

Show that $\lim_{n \rightarrow +\infty} x_n$ exists and calculate the limit.

SCHOOL OF MATHEMATICAL SCIENCES, PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: 2201110024@stu.pku.edu.cn